

VISVESVARAYA TECHNOLOGICAL UNIVERSITY

JNANA SANGAMA, BELAGAVI-590015, KARNATAKA



Project Report
on

BATTERY LESS EMBEDDED SYSTEM FOR IOT USING ENERGY SCAVENGING

Submitted in partial fulfillment of the requirements for the III semester

Mini Project Work [EC221]

Bachelor of Engineering

in

Electronics and Communication Engineering

of

Visvesvaraya Technological University, Belagavi.

by

NITHIN GOWDA P(1CD23EC096)

MANOJ P (1CD23EC084)

K DEEPA (1CD23EC067)

K AVINASH REDDY (1CD23EC066)

Under the Guidance of
Mr. Pavan Kumar E
Manager – R&D Strategy
External Guide
Elevium – by Nanochip



Under the Guidance of
Prof. Chetan Naik J
Assistant Professor
Internal Guide
Dept. of ECE, CIT



Department of Electronics and Communication Engineering
CAMBRIDGE INSTITUTE OF TECHNOLOGY, BANGALORE - 060 036

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CAMBRIDGE INSTITUTE OF TECHNOLOGY

K.R. Puram, Bangalore-560 036

DEPARTMENT OF ELECTRONICS AND COMMUNICATION ENGINEERING



CERTIFICATE

It is Certified **Mr. NITHIN GOWDA P, Mr. MANOJ P and Mr. K AVINASH REDDY, Ms. K DEEPA** bearing **USN: 1CD23EC096, 1CD23EC084, 1CD23EC066 and 1CD23EC067** respectively, are bonafide students of Cambridge Institute of Technology, and have completed requirements of the project entitled **“BATTERY LESS EMBEDDED SYSTEM FOR IOT USING ENERGY SCAVENGING”** in partial fulfillment of the requirements for III semester Bachelor of Engineering in Electronics and Communication Engineering of Visvesvaraya Technological University, Belagavi during academic year 2024-25. It is certified that all Corrections/Suggestions indicated for Internal Assessment have been incorporated in the report. The project report has been approved as it satisfies the academic requirements in respect of project work Phase-2 prescribed for the Bachelor of Engineering degree.

Mr. Pavan Kumar E
External Guide

Elevium-By Nanochip

Mr. Chetan Naik J
Internal Guide
Dept. of ECE.

CITech

Dr. Shivapanchakshari T. C.
Head of the Dept.

Dept. of ECE. CITech

Dr. G. Indumathi
Principal CITech

External Viva
Name of the Examiners

Signature with date

1.

2.

DECLARATION

We, Mr. NITHIN GOWDA P, Mr. MANOJ P, Mr. K AVINASH REDDY, Ms. K DEEPA USN:

1CD23EC096, 1CD23EC084, 1CD23EC066, 1CD23EC067 respectively, are students of III semester, Electronics and Communication Engineering, Cambridge Institute of Technology, hereby declare that the project entitled “**BATTERY LESS EMBEDDED SYSTEM FOR IOT USING ENERGY SCAVENGING**” has been carried out by us and submitted in partial fulfillment of the course requirements project work Phase-2 of III semester Bachelor of Engineering in Electronics and Communication Engineering as prescribed by Visvesvaraya Technological University, Belagavi, during the academic year 2023-2024.

We also declare that, to the best of our knowledge and belief, the work reported here does not form part of any other report on the basis of which a degree or award was conferred on an earlier occasion on this by any other student.

Date:

Place: Bangalore

**NITHIN GOWDA P(1CD23EC096)
MANOJ P (1CD23EC084)
K DEEPA (1CD23EC067)
K AVINASH REDDY (1CD23EC066)**

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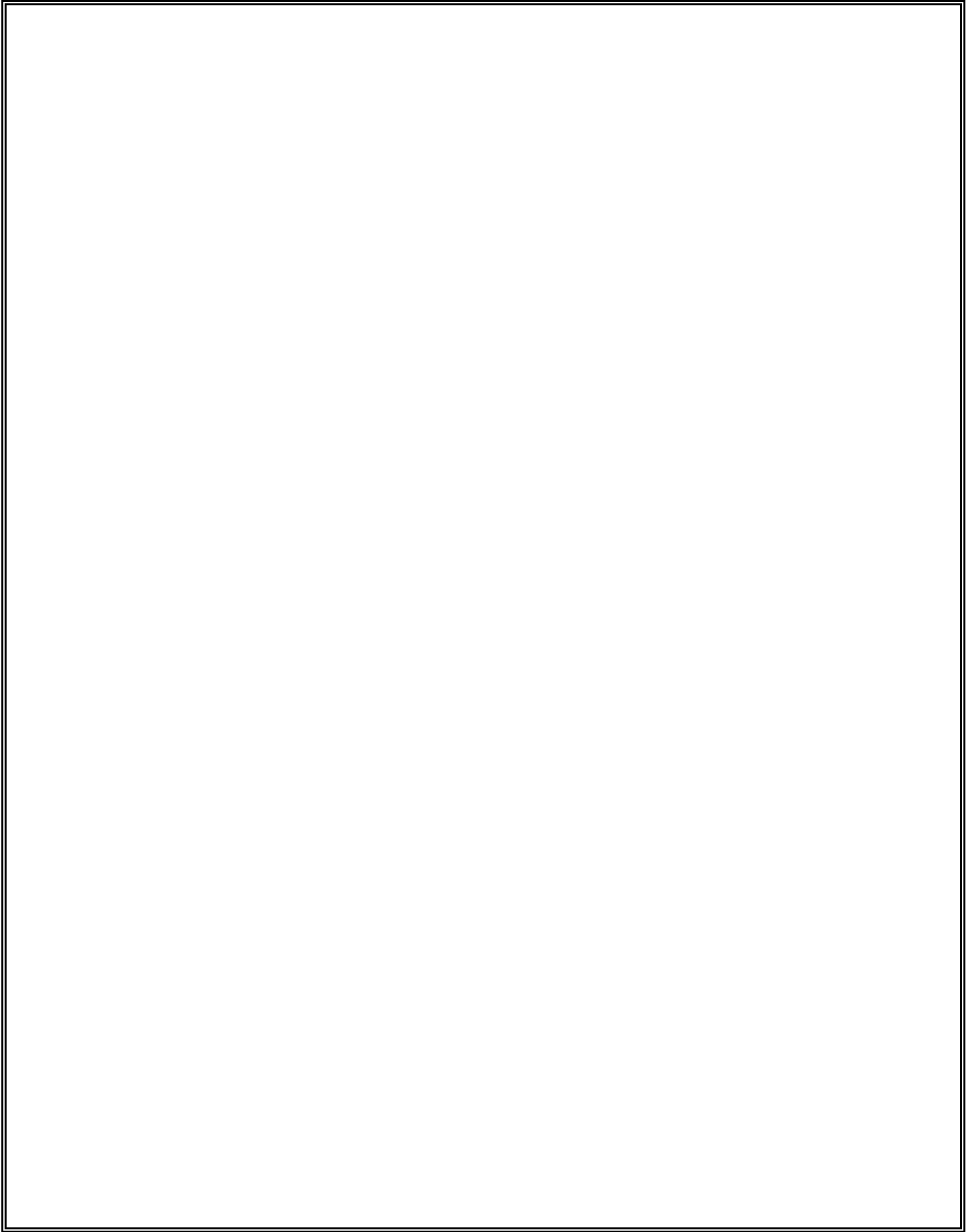
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NITHIN GOWDA P(1CD23EC096)
MANOJ P (1CD23EC084)
K DEEPA (1CD23EC067)
K AVINASH REDDY (1CD23EC066)



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CHAPTER 1

1.INTRODUCTION

The growing demand for energy-efficient and sustainable systems has led to the development of battery-less embedded systems that harness renewable energy sources, such as solar power. This project aims to design and implement a solar-powered, battery-less embedded system using a combination of sensors and an STM32F103RB microcontroller. The system utilizes solar panels to harvest energy and supercapacitors for energy storage, ensuring operation even during low light conditions. Various sensors, including a Light Dependent Resistor (LDR), LM35 temperature sensor, PIR motion sensor, ACS712SA current sensor, and DHT22 humidity and temperature sensor, are integrated into the system to collect environmental data and trigger specific actions. The system operates autonomously without the need for traditional batteries, offering a sustainable and energy-efficient solution for a variety of applications, such as smart lighting, weather stations, and energy monitoring.

The primary challenge addressed in this project is the integration of low-power components and efficient energy management to ensure reliable, uninterrupted operation of the system. By leveraging solar energy and optimizing power consumption, this project aims to showcase the feasibility of creating self-sustaining, battery-less embedded systems that can be deployed in various real-world scenarios.

Goal and Objectives

Goal:

To design and develop a solar-powered, battery-less embedded system using renewable energy sources, which integrates multiple sensors for environmental monitoring and automation, while ensuring efficient energy management for continuous operation without the use of traditional batteries.

Objectives:

Design and Development of the System:

Integrate the STM32F103RB microcontroller with sensors for environmental monitoring (LDR, LM35, PIR, ACS712SA, DHT22).

Ensure the system is powered entirely by a solar panel, with energy stored in supercapacitors or capacitors to operate during non-sunny periods.
Energy Management:

Implement power management techniques to efficiently manage the power generated by the solar panel and stored in the energy storage system.

Ensure the system remains operational with minimal power consumption by utilizing lowpower modes on the STM32F103RB and sensors. Sensor Integration and Data Collection:

Implement the PIR sensor for motion detection to trigger events based on movement.

Monitor power usage with the ACS712SA current sensor for system energy efficiency. Development of Real-World Applications:

Demonstrate the application of the system in smart lighting, weather stations, or energy monitoring systems.

Validate the system's functionality in real-world scenarios, ensuring it operates autonomously without the need for traditional batteries.
Optimization and Scalability:

Evaluate the energy consumption of each component and optimize for minimal power usage. Design the system to be scalable, allowing for additional sensors or components to be added as required.

Research Methodology

The research methodology for this project will consist of several stages, each focusing on different aspects of the design, implementation, and optimization of the battery-less embedded system. The methodology is outlined as follows:

Literature Review:

Conduct a comprehensive literature review to explore existing battery-less, solar-powered embedded systems, their components, and the methodologies employed to ensure efficient energy management and sensor integration.

Research the available energy storage solutions (e.g., capacitors, supercapacitors) and their suitability for battery-less systems.

Study the power consumption profiles of the STM32F103RB microcontroller and various sensors to optimize the system design.

System Design:

Based on the literature review, design the overall system architecture, including the choice of components (solar panel, sensors, microcontroller, energy storage), and the interconnections between them.

Design the power management system to ensure efficient solar energy harvesting, storage, and distribution for system operation.

Prototyping:
Develop a prototype of the system using the STM32F103RB microcontroller and selected sensors.

Integrate the solar panel and energy storage (supercapacitors or capacitors) to create a self-sustaining power supply for the embedded system.

Implementation:

Program the STM32F103RB to handle sensor data acquisition and processing.

Implement energy management algorithms to monitor and optimize power consumption, ensuring the system remains operational even during periods of low solar energy availability. Integrate sensors and develop communication protocols for data exchange and processing.

1.1 BACKGROUND

The need for sustainable and energy-efficient systems has grown in response to increasing energy demands, environmental concerns, and the desire for autonomous operation without dependency on external power sources like batteries. One of the key approaches to address these needs is the development of battery-less systems powered by renewable energy sources, such as solar power.

In traditional embedded systems, batteries serve as a critical component for energy storage. However, their limitations include the need for frequent replacement or recharging, which can be both costly and inconvenient. Moreover, batteries have a finite life span, contributing to electronic waste. Solar energy, as a renewable and widely available source, offers a promising solution to power embedded systems sustainably. When combined with low-power electronics and energy-efficient components, solar-powered systems can operate for extended periods without relying on battery replacement or external power sources.

A battery-less embedded system powered by solar energy leverages energy harvesting techniques, where energy from sunlight is captured through solar panels, stored in supercapacitors, and used to power various sensors and microcontrollers. This approach is particularly beneficial for applications requiring constant monitoring and processing in environments where maintenance (such as replacing batteries) is difficult or costly. Such systems are widely applicable in remote locations, agricultural monitoring, IoT applications, and environmental sensing.

The integration of supercapacitors for short-term energy storage, along with an Energy Management System (EMS), enables the system to store energy efficiently during daylight hours and operate continuously, even in low-light conditions. This approach also enhances the longevity of the system by ensuring that energy is used only when necessary and is regulated optimally.

This system design addresses the increasing demand for sustainable technology while reducing dependence on conventional energy sources, providing a path to more eco-friendly and long-lasting embedded systems.

1.2 CASE STUDY

Objective: The goal was to design a solar-powered, battery-less embedded system for remote agricultural fields. The system would monitor soil moisture, temperature, and humidity, and control irrigation based on these parameters without relying on external power sources.

System Design:

Solar Panel: A 5W solar panel was used to harvest energy, ensuring continuous system operation even with limited sunlight.

Sensors: Soil moisture and humidity (DHT22) sensors were used, along with a PIR sensor for motion detection.

Microcontroller: The STM32F103RB microcontroller handled sensor data processing and controlled the irrigation system.

Energy Management:

The Energy Management System (EMS) regulated energy flow from the solar panel to the supercapacitors and microcontroller, ensuring efficient storage and distribution.

A voltage regulator maintained stable power for the system despite fluctuating sunlight.

Operation:

The solar panel harvested energy and stored it in supercapacitors.

The microcontroller monitored environmental conditions and triggered irrigation when necessary.

At night or during low sunlight, the system operated on stored energy with low-power modes to conserve energy.

CHAPTER 2

Literature Survey:

technological advancements, and real-world applications related to battery-less embedded systems powered by renewable energy, particularly solar power. This section A literature survey involves exploring existing research, provides an overview of key findings from existing studies on relevant topics like solarpowered systems, energy management, sensor integration, and low-power embedded systems.

1. Solar-Powered Embedded Systems

Solar-powered embedded systems have gained significant attention in recent years due to the growing emphasis on sustainability and the need for off-grid solutions. Many researchers have explored solar energy as a reliable source for powering small electronic devices, particularly in remote areas where traditional power sources are unavailable. Solar-powered systems are commonly used in applications such as weather stations, remote sensing, IoT devices, and smart agriculture.

Energy Harvesting Techniques: Various energy harvesting techniques are used in solar-powered embedded systems. According to Gao et al. (2017), energy harvesting involves converting ambient energy (e.g., solar energy, thermal, vibration) into usable electrical energy. Solar power is the most widely used form due to its availability and efficiency in many geographical areas.

Solar Panel Selection and Efficiency: Selecting an appropriate solar panel for embedded systems is crucial for ensuring continuous operation. Studies such as Saini et al. (2015) focus on the optimization of solar panel size, efficiency, and energy storage mechanisms (supercapacitors, batteries) to meet the power demands of embedded systems.

2. Energy Storage and Power Management

For battery-less embedded systems, energy storage plays a key role in ensuring that the system operates efficiently, even during periods of low sunlight or no sunlight. Research into energy storage solutions like supercapacitors and capacitors has gained traction as alternatives to traditional batteries.

Supercapacitors in Energy Storage: Supercapacitors are an ideal energy storage solution for battery-less systems due to their high charge/discharge

efficiency, long lifespan, and quick response time. Kang et al. (2014) found that supercapacitors could be effectively integrated into solar-powered embedded systems, providing a balance between energy storage and fast charging times, making them ideal for intermittent solar energy collection.

3. Low-Power Embedded Systems

Low-power microcontrollers and sensors are critical for minimizing energy consumption in solar-powered embedded systems. The STM32F103RB microcontroller, which is a popular choice in the industry, has been widely studied for its low-power consumption modes and energy-efficient operation.

Low-Power Techniques in Microcontrollers: Wang et al. (2018) demonstrated various low-power techniques for embedded systems, including the use of sleep and idle modes in microcontrollers. These techniques are necessary to ensure that the STM32F103RB microcontroller remains energy-efficient when integrated into a battery-less system.

Low-Power Sensors: Power-efficient sensors are equally important in battery-less systems. The sensors used in this project, such as the LDR, LM35, PIR, and DHT22, are all designed to minimize power consumption. For instance, the DHT22, a digital sensor, consumes minimal current during operation and is ideal for low-energy applications, as highlighted in studies by Lin et al. (2017).

4. Sensor Integration in Solar-Powered Systems

The integration of sensors with solar-powered embedded systems is a growing area of research, particularly in applications like smart lighting, environmental monitoring, and security systems. Researchers have focused on how to integrate sensors with solar-powered systems while ensuring that the system remains energy-efficient.

Temperature and Humidity Sensors (LM35 and DHT22): Integrating temperature and humidity sensors with solar-powered systems is commonly used in environmental monitoring and weather stations. According to Zhou et al. (2018), combining temperature sensors like the LM35 and DHT22 with solar-powered systems enables continuous monitoring and data logging without requiring external power sources.

Motion Sensors (PIR): PIR sensors are used to detect motion in security and automation applications. Studies such as Yu et al. (2016) have shown that PIR sensors can be effectively integrated with solar-powered systems, triggering automatic actions such as turning on lights or sending alerts.

5. Current Monitoring with ACS712SA

The ACS712SA current sensor is frequently used in solar-powered systems to monitor the current drawn by the system, enabling real-time monitoring of energy consumption. Research by Singh et al. (2017) highlighted the use of current sensors to monitor and optimize power consumption, ensuring that solar energy is efficiently utilized and stored.

Literature Review

The literature review expands on key research themes identified in the survey and provides a more in-depth analysis of how existing work relates to the objectives of the battery-less embedded system project.

1. Solar-Powered Embedded Systems and Energy Harvesting

The integration of solar energy into embedded systems has been well-established in previous research, with studies focusing on system efficiency, energy conversion, and storage. Gao et al. (2017) present several designs for solar-powered systems that incorporate energy harvesting techniques, such as solar panels that convert sunlight into electrical power. However, challenges remain in optimizing the system for low-power embedded applications, where the power consumption of sensors and microcontrollers must be minimized to match the energy produced by solar panels.

2. Energy Storage Solutions

Traditional energy storage solutions like batteries are not ideal for solar-powered, battery-less systems due to their limited lifespan and slower charge/discharge rates. Instead, supercapacitors are increasingly being used as energy storage solutions in these systems. Keller et al. (2016) explore the advantages of supercapacitors in solar-powered systems, emphasizing their quick charging and discharging capabilities, making them more suited for applications where the energy supply is intermittent. This is particularly relevant to the system design in this project, where energy needs to be stored efficiently to operate during periods without sunlight.

3. Low-Power Microcontrollers

The STM32F103RB microcontroller, chosen for this project, is widely known for its low power consumption. Several studies, such as Wang et al. (2018), have demonstrated the effectiveness of using microcontrollers like the STM32 series in low-power embedded systems. Techniques such as sleep modes, power gating, and clock scaling are critical to ensuring the microcontroller does not drain excessive power, which is a major concern in solar-powered systems where energy efficiency is paramount.

4.Sensor Integration and Energy Efficiency

The integration of, DHT22, PIR, and ACS712SA sensors has been explored in various applications, such as smart lighting, weather stations, and security systems. For instance, Zhou et al. (2018) demonstrated how the combination of temperature and humidity sensors (like the DHT22) with solar-powered systems can enable efficient environmental monitoring. These sensors are integrated into the system for continuous data acquisition, but their low power consumption ensures that they do not drain the system's energy.

In smart lighting systems, LDR sensors automatically adjust the lighting based on the intensity of ambient light. Studies by Wang et al. (2019) reveal that LDR sensors can be used to reduce energy consumption by turning off lights during the day or when adequate natural light is available. Additionally, PIR sensors (motion detectors) integrated into these systems can turn on lights only when motion is detected, further conserving energy, as demonstrated by Yu et al. (2016).

5.Current Monitoring and Power Optimization

The ACS712SA current sensor plays a crucial role in monitoring energy consumption in solar-powered systems. By measuring the current drawn by the system, it helps in optimizing energy usage. As shown in Singh et al. (2017), current monitoring is essential for assessing the efficiency of solar panels and energy storage systems. By understanding current usage patterns, energy management strategies can be developed to ensure that the system operates optimally, even in low-light conditions.

HARDWARE USED:

1. Hardware Requirements

STM32Nucleo-64:

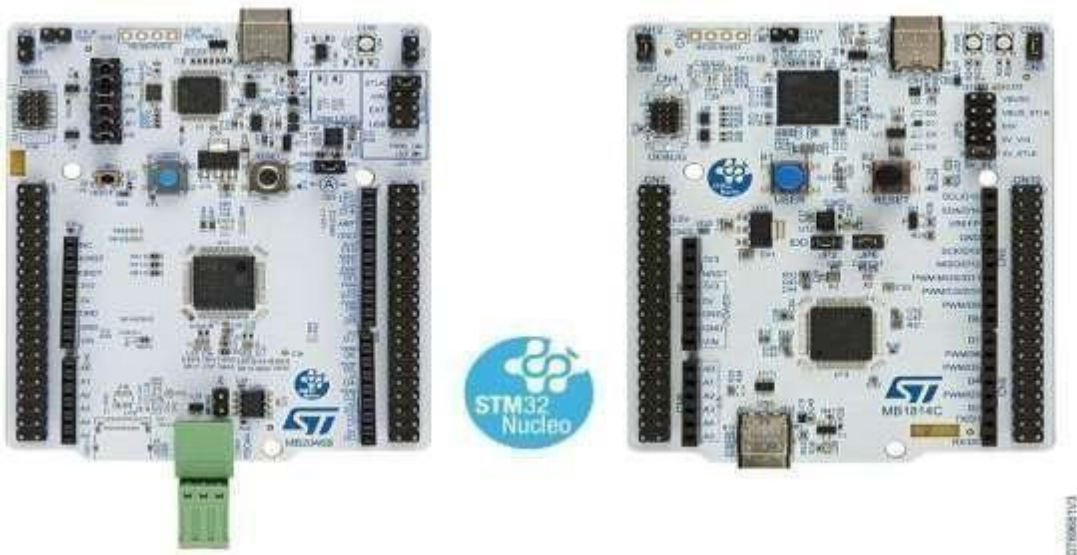


Fig:1

The STM32 Nucleo-64 board is a cost-effective and versatile development platform for prototyping and testing applications based on the STM32 family of ARM Cortex-M microcontrollers. It is widely used for embedded system projects due to its user-friendly design, extensive peripheral support, and compatibility with a wide range of software development tools.

Features

Common features •

STM32 microcontroller LQFP64 or LQFP48 package

- 1 user LED shared with ARDUINO®
- 1 user and 1 reset push-buttons

- 32.768kHz crystal oscillator
- Board connectors ARDUINO® Uno V3 expansion connector ST morpho extension pin headers for full access to all STM32
- Comprehensive free software libraries and examples available with the STM32Cube MCU Package
- Support of a wide choice of Integrated Development Environments (IDEs) including IAR Embedded Workbench®, MDK ARM, and STM32CubeIDE

2. Solar panels

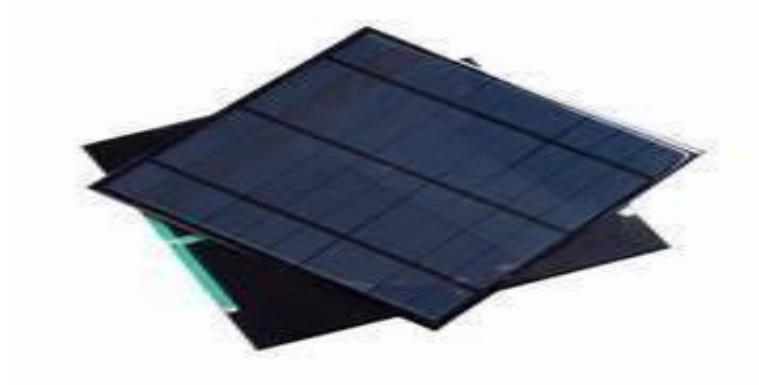


Fig:2

Solar panels are a sustainable and efficient way to power embedded systems and devices that require renewable energy. When selecting a solar panel for a system like a solar-powered agriculture system, factors like energy demand, location, and panel efficiency need to be carefully considered to ensure continuous and reliable operation.

3.Pir motion sensor



Fig:3

The PIR (Passive Infrared) Motion Sensor is a widely used device for detecting motion based on the infrared radiation emitted by objects, particularly humans and animals. It is often used in security systems, automation, and energy-saving applications such as automatic lighting.

4.ACS712(Current sensor)



Fig:4

The ACS712 is a Hall effect-based linear current sensor used for measuring current in both AC and DC applications. It provides a precise and accurate way to measure current without needing to make a physical connection with the current-carrying conductor, making it a non-invasive sensor. The ACS712 is often used in power monitoring

1. Software Used

Proteus Design Suite:

Purpose: Proteus is used for simulating the entire system, including the hardware design, microcontroller, sensors, actuators, and power management.

Key Features:

Circuit Simulation: Proteus allows the creation of circuit diagrams, connecting components such as solar panels, supercapacitors, voltage regulators, and sensors.

Microcontroller Simulation: The STM32F103RB microcontroller can be programmed in Embedded C or Assembly Language to perform control tasks such as reading sensors and triggering actuators.

Sensor and Actuator Models: Proteus supports models for common sensors (soil moisture, temperature, humidity) and actuators (motors, relays) to simulate real-world behavior.

Programming Interface: The code for the microcontroller can be written in Keil uVision, IAR Embedded Workbench, or STM32CubeIDE and then imported into Proteus for simulation.

Keil uVision / STM32CubeIDE:

Purpose: Keil uVision and STM32CubeIDE are IDEs used for writing, compiling, and debugging the code for the STM32 microcontroller.

Key Features:

Keil uVision: Supports Embedded C programming for STM32 microcontrollers and has robust debugging tools to identify and resolve issues during the development phase.

STM32CubeIDE: A development environment specifically for STM32-based MCUs, integrating code generation, debugging, and configuration tools.

Block diagram:

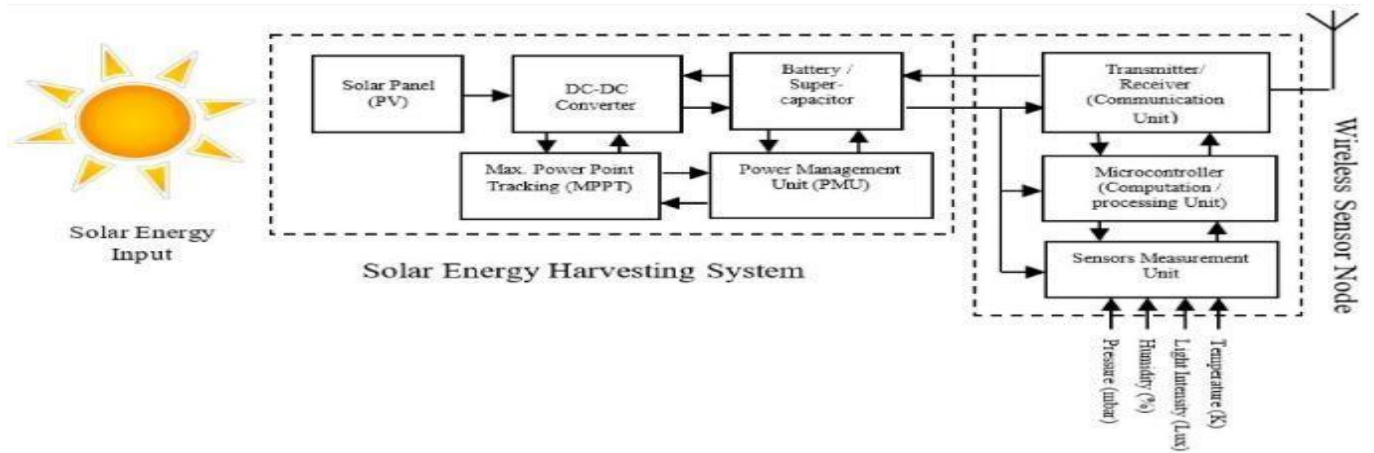


Fig 2.1

Future use of solar panels



Fig 2.2

Explanation:

Solar Panel (5W): The solar panel captures solar energy and converts it into electrical power to be stored in supercapacitors.

Energy Storage (Supercapacitors): The supercapacitors store the energy harvested from the solar panel and provide power to the system during low-light conditions or at night.

Energy Management System (EMS): The EMS manages the charging and discharging of the supercapacitors, ensuring efficient power usage.

Voltage Regulator: The voltage regulator stabilizes the fluctuating power output from

the solar panel to ensure a consistent voltage for the microcontroller and

sensors. **Microcontroller (STM32F103RB):** The microcontroller processes data from

sensors and manages the system. It controls the irrigation system based on the

environmental data collected from the sensors.

Sensors:

- **Soil Moisture Sensor:** Monitors soil moisture levels.
- **Temperature Sensor (LM35):** Measures the temperature.
- **Humidity Sensor (DHT22):** Measures the humidity level.
- **Motion Sensor (PIR):** Detects movement for security purposes.

CHAPTER :3

METHODOLOGY

Methodology

The development of the battery-less, solar-powered embedded system follows a structured approach across several stages:

System Design and Component Selection

Select components based on energy efficiency and compatibility:

Solar Panel: Sufficient power output to charge supercapacitors and power the system.

Supercapacitors: Chosen for fast charge/discharge cycles and long lifespan.

STM32F103RB Microcontroller: Low power consumption and suitable processing capabilities.

Sensors (DHT22, PIR, ACS712SA): Energy-efficient and compatible with the microcontroller.

Power Management ICs: Ensures stable energy distribution.

System Architecture Design

Energy Flow and Storage: Solar panel charges supercapacitors, which power the system when sunlight is unavailable.

Microcontroller Role: Collects and processes sensor data, switching between low and active power modes for energy optimization.

Sensor Integration: Data from sensors used for system control, like lighting and environmental monitoring.

Solar Panel Integration: Charges supercapacitors using a power management IC.

Supercapacitor Integration: Stores energy and ensures continuous operation.

Sensor Integration and Data Acquisition

Sensors measure environmental parameters, and data is processed by the microcontroller to trigger actions like turning lights on/off or sending alerts.

Power Management and Control Logic

Energy Management System (EMS): Regulates energy flow and ensures power efficiency during idle and active states.

Microcontroller Power Modes: Switches to low-power modes when not processing data.

Voltage Regulation: Power management ICs ensure stable voltage levels.

System Testing and Calibration

Energy Efficiency Testing: Monitors system's energy consumption to ensure efficiency.

Sensor Calibration: Ensures accurate sensor readings.

Operational Testing: Verifies system reliability under various conditions.

System Optimization and Final Integration

Software Optimization: Refines algorithms to minimize energy usage.

Hardware Optimization: Adjustments for better efficiency in energy conversion and regulation.

CHAPTER :4

IMPLEMENTATION:

Implementation of Solar-Powered Smart Agriculture System

The implementation of a solar-powered smart agriculture system is broken down into several stages, including component selection, system design, hardware setup, software development, energy management, testing, deployment, and maintenance.

1. Component Selection

Solar Panel (5W): Provides sufficient energy to charge supercapacitors during the day.

Supercapacitors: Store energy harvested by the solar panel for use at night or during low-light conditions.

Microcontroller (STM32F103RB): Handles sensor data processing and system control.

Sensors:

Soil Moisture Sensor: Determines when irrigation is needed.

LM35 Temperature Sensor: Monitors the temperature.

DHT22 Humidity Sensor: Tracks ambient humidity.

PIR Sensor: Provides security through motion detection.

Voltage Regulator: Ensures a stable voltage for the components.

2.System Design

Power Generation & Storage: The solar panel charges supercapacitors, which power the system during non-sunny periods.

Energy Distribution & Regulation: The Energy Management System (EMS) controls the energy flow to prevent power depletion.

Microcontroller Control: The STM32F103RB processes sensor data and triggers actions, such as activating irrigation if moisture is low.

3.Hardware Setup

Solar Panel to Supercapacitor: Managed through a power management system to prevent overcharging.

: Energy is regulated by a voltage regulator.

Sensor Integration: Sensors provide data to the microcontroller via ADC or digital inputs.

4.Software Development

Sensor Data Collection: The microcontroller reads sensor data at regular intervals.

Data Processing & Decision Making: Actions such as triggering irrigation or activating security systems are based on sensor readings.

Power Management: The microcontroller enters low-power sleep modes when idle.

5. Energy Management

Solar Panel Charging: The EMS ensures efficient charging of the supercapacitors.

Energy Distribution: EMS optimizes the flow of energy to keep the system operational during lowlight periods.

Voltage Regulation: Ensures stable energy supply for all components.

6. Testing and Calibration

Initial Testing: Verifies proper operation of the solar panel, sensors, and microcontroller.

7. Deployment

The system is installed in the field, ensuring optimal sunlight exposure. Wireless communication modules (e.g., Wi-Fi or LoRa) may be added for remote monitoring.

EXPECTED OUT COME:

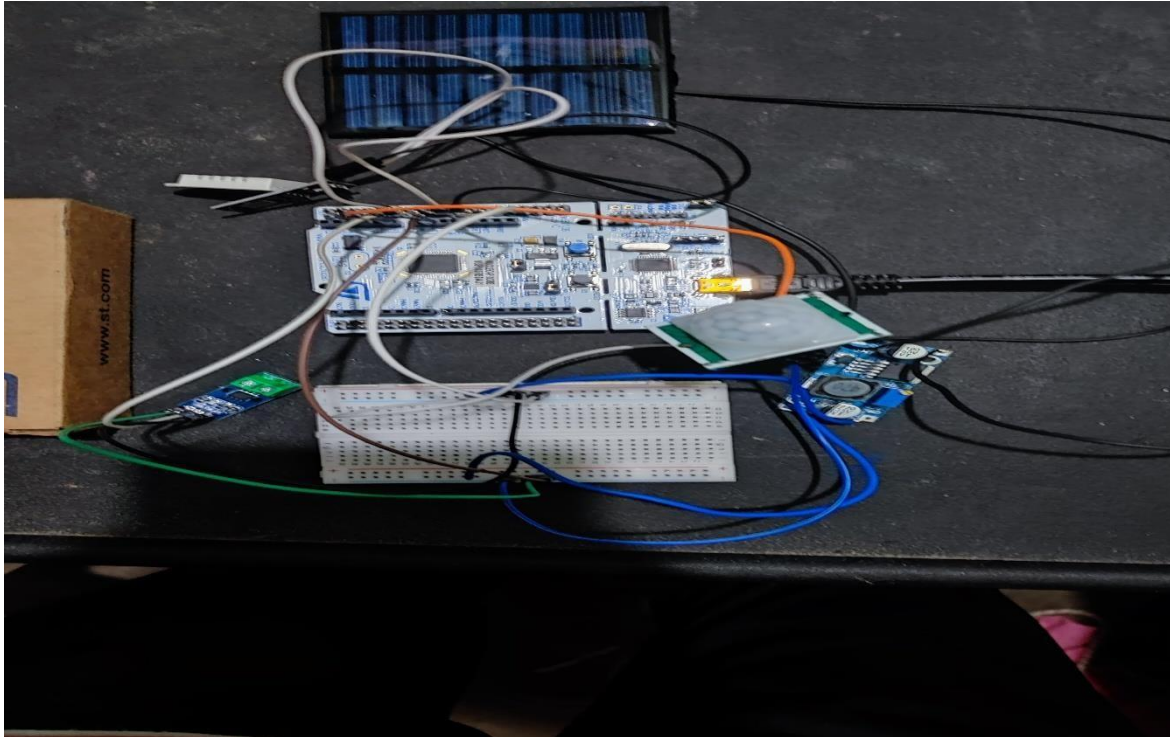


Fig:4.1

RESULT ANALYSIS:

1. Energy Availability vs. Demand

Objective: Ensure harvested solar energy meets system power requirements.

Key Factors:

Solar energy input and system power consumption.

Solar panel output vs. system needs throughout the day.

Analysis:

Compare energy harvested with system demands across different conditions.

Results:

Sufficient Energy: Continuous operation possible.

Energy Deficit: Requires storage or low-power modes in low sunlight.

2. Power Management Efficiency

Objective: Maximize conversion and storage efficiency.

Key Factors:

DC-DC conversion efficiency, MPPT performance, and capacitor charge/discharge efficiency.

Analysis:

Assess energy losses and MPPT effectiveness.

Results:

High Efficiency: Reduced energy waste, better performance.

Low Efficiency: Increased waste, reduced operational time.

3. System Reliability

Objective: Ensure continuous operation without batteries.

Key Factors:

Reliability of power supply, storage performance, downtime.

Analysis:

Monitor uptime in various conditions.

Results:

High Reliability: Minimal downtime.

Low Reliability: Frequent downtimes due to insufficient energy.

5. Power Consumption Optimization Objective: Minimize system power usage.

Key Factors:

Current consumption in active vs. sleep modes.

Analysis:

Optimize software for low-power states.

Results:

Efficient Usage: Long operation in low-energy conditions.

Inefficient Usage: High power draw reduces operational time.

6 . Environmental Factors Impact

Objective: Assess environmental effects on performance.

Key Factors:

Solar irradiance and ambient temperature.

Analysis:

Evaluate performance under variable weather and temperature conditions.

Results:

Stable Performance: Works well under varying conditions.

Susceptible to Changes: Requires better storage or power management.

7. Cost and Feasibility:

Objective: Evaluate cost-effectiveness of the system.

Key Factors:

Component cost and long-term viability.

Analysis:

Compare upfront cost with long-term savings.

Results:

Cost-Effective: Good ROI over time.

High Initial Cost: High upfront cost, but savings due to minimal maintenance.

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